

Discovery Plane Theory in Practice

A Complete Worked Guide to Scientific Question Evaluation, Discovery Complexity, and Question Compression

A Fully Worked University Energy-Research Example

Purpose. Every research project begins with a question, yet few researchers are formally taught how to evaluate whether one question is scientifically stronger than another. This guide translates the formal framework of Discovery Plane Theory into a transparent procedure for teaching, research-question evaluation, and research screening. Every principal calculation is presented manually, explained step by step, and linked directly to its formal mathematical basis.

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Preface: Why This Guide Was Written

For centuries, scientific progress has been described through discoveries, theories, and answers. Yet every discovery begins much earlier—with a question. Long before data are collected, experiments are designed, or hypotheses are tested, a researcher must decide which question is worth pursuing. That decision often determines the value of everything that follows.

During research supervision and academic discussions, I repeatedly observed that students and even experienced researchers could explain how to conduct research, yet many found it difficult to justify why one research question should be preferred over another. Considerable effort was often invested in answering questions whose scientific potential had never been examined systematically. This suggested that research methodology devotes far more attention to answering questions than to evaluating the questions themselves.

Discovery Plane Theory (DPT) was developed to address this earlier stage of scientific reasoning. Rather than replacing expert judgement or disciplinary knowledge, it provides a transparent mathematical framework for representing, comparing, and discussing research questions through explicitly declared assumptions and reproducible calculations.

This guide was written to bridge the gap between the formal theory and its practical application. It does not replace the theoretical framework or the computational benchmark. Instead, it demonstrates every principal calculation through a complete worked example in which every assumption, score, and intermediate result is declared openly and can be verified manually.

The example intentionally uses university electricity consumption so that readers from engineering, computer science, management, medicine, education, law, the social sciences, and the humanities can concentrate on understanding the workflow rather than specialised disciplinary knowledge. Readers are encouraged to replace the example with problems from their own fields while preserving the same sequence of reasoning.

The central teaching principle of this manual is simple: no numerical result should appear without a declared basis. Reality Space, Background Knowledge, admissible procedures, cost functional, description language, normalisation, thresholds, weights, and uncertainty assumptions are therefore introduced before they are used. The goal is not merely to calculate Discovery Plane coordinates, but to help readers understand why each calculation is necessary and how it contributes to a transparent evaluation of scientific questions.

Key idea

This guide presents a fully worked educational example. All numerical values, scores, thresholds, and calculations are transparent pedagogical conventions used solely to illustrate the application of Discovery Plane Theory (DPT). They are not official re-evaluations of the cited studies and should not be interpreted as proof that a selected research question will necessarily lead to a scientific discovery.

Glossary of Core Terms

Term	Meaning in this guide
Reality Space	The declared observations, measurements, anomalies, regularities, or articulated facts from which candidate questions are considered.
Background knowledge	The theories, skills, tools, data, and institutional capabilities already available to the research team.
Discovery procedure	A finite cognitive, computational, inferential, experimental, or collaborative process capable of generating a question from the declared Reality Space and background knowledge.
Discovery Complexity (DC)	The least admissible cost of making a question available under the declared conventions.
Question Compression (QC)	The nonnegative reduction in representational description length produced when a question reorganises the Reality Space.
Discovery coordinate	The ordered pair containing a question's DC and QC values.
Threshold	A declared analytical boundary used to describe the four Discovery Plane regions. It is a convention, not a natural law.
Dominance	A comparison in which one question has no greater DC and no smaller QC than another, with at least one strict improvement.
Discovery Frontier	The set of nondominated candidate questions.
Weighted distance	A geometric distance between two Discovery coordinates under declared positive weights.
Neighbourhood	The set of questions lying within a declared distance radius of a question.
Geodesic	A shortest admissible path between questions in a connected question-transition graph.
Regularised utility	A scalar summary that rewards QC and penalises DC according to a declared parameter.
Representation	The formal language, graph, set of propositions, model, or other coding system used to describe the Reality Space and its transformation.

Abstract

Research training commonly teaches students how to review literature, collect data, test hypotheses, and report findings, but it gives less formal attention to the step that logically precedes all of them: selecting a scientifically productive question. Discovery Plane Theory represents a question through two coordinates: Discovery Complexity, the least admissible cost of making the question available from a declared Reality Space and background knowledge, and Question Compression, the reduction in representational description length induced by that question. This manual develops a complete worked example based on university electricity consumption. Four candidate questions are generated, scored, normalised, classified into Discovery Plane regions, compared through dominance, screened through the Discovery Frontier, evaluated under a research budget, analysed through distance and utility, and tested for threshold, weight, and error sensitivity. Each step states its purpose, formula, manual calculation, interpretation, and limitation. The final result demonstrates how DPT can convert an intuitive statement such as “this appears to be a good research question” into a transparent and reproducible research-screening argument.

Keywords: Discovery Plane Theory; research question formation; Discovery Complexity; Question Compression; Discovery Frontier; research methodology; worked example; reproducible screening.

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1 How to Use This Manual

Every research project begins with a question. Yet surprisingly little formal guidance exists for deciding whether one scientific question is likely to produce deeper understanding than another before years of work are invested. The broader historical and conceptual motivation for viewing scientific progress through the questions that shape discovery, rather than only through the answers they produce, is explored in *The Mystery of Questions: Why Some Questions Change the World* [1]. Discovery Plane Theory extends this perspective by providing a formal mathematical framework for representing, evaluating, and comparing scientific research questions [2], while its computational implementation and empirical evaluation are presented in the accompanying benchmark study [3].

This guide demonstrates one complete application of Discovery Plane Theory (DPT) through a fully worked example. Rather than presenting only definitions or theorems, every major calculation is performed manually so that each intermediate result can be verified independently. The objective is not merely to obtain a final score, but to understand why every quantity is calculated, how it is derived, and how it contributes to transparent and reproducible research-question evaluation.

The document is designed to be read sequentially because every later calculation depends upon declarations introduced earlier. Skipping steps may change the interpretation of subsequent results.

Each numbered step contains five components:

1. what must be declared or calculated;
2. why the step is necessary;
3. the governing DPT expression;
4. a complete manual calculation; and
5. an interpretation in plain language.

Throughout the guide, three different layers are kept strictly separate:

- (a) **Formal DPT quantities:** the mathematical definitions of DC, QC, coordinates, dominance, distance, utility, and dynamic evolution;
- (b) **Operational conventions:** the declared scoring rules, description language, admissible procedures, thresholds, weights, and normalisation used for the worked example;
- (c) **Empirical evidence:** observations, measurements, experiments, coding procedures, and scientific conclusions obtained from real investigations.

This distinction is essential because Discovery Plane Theory evaluates scientific questions relative to explicitly declared conventions. Discovery Complexity is therefore not an intrinsic property of a sentence; it depends on the declared Reality Space, background knowledge, admissible procedures, cost functional, and representational language [2]. Likewise, the computational benchmark operationalised these formal quantities using transparent proxy components suitable for reproducible evaluation rather than claiming direct measurement of the theoretical constructs [3].

Key idea

Do not begin by assigning scores. Begin by declaring what is observed, what is already known, and which alternative questions are genuinely available. Every later calculation depends on these first declarations.

Important limitation

A high DPT score does not prove that a research question is true, ethical, feasible, fundable, publishable, or historically important. Discovery Plane Theory provides a transparent framework for comparing candidate questions. Final research decisions continue to require expert judgement, empirical evidence, ethical review, available resources, and scientific validation.

Try it yourself

Before reading further, choose one research topic from your own discipline. Write down the observations that motivated it, describe the knowledge already available, and formulate at least three different candidate research questions. Repeat the exercise after completing the guide and compare the two versions.

2 The Central Research Problem

The practical problem

Universities often have many observations but no single question that organises them. Faculty and students may begin with a convenient method, a fashionable technology, or an easily available dataset, and only afterwards construct a question. DPT reverses this order: first declare the evidence and knowledge state, then generate and compare candidate questions before committing major research resources.

The central DPT thesis is that scientifically admissible questions can be represented, relative to declared conventions, in a two-dimensional Discovery Plane whose coordinates are Discovery Complexity and Question Compression [2]. The complete teaching sequence followed in this guide is summarised in fig. 1.

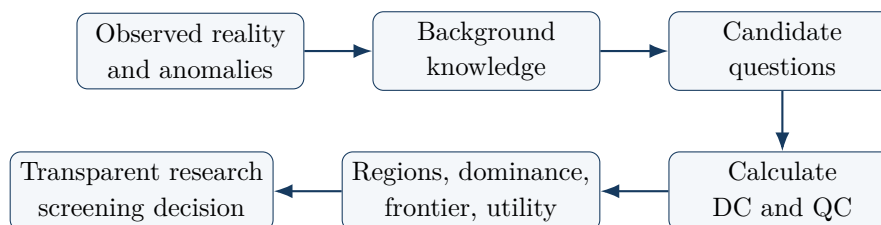


Figure 1: The teaching workflow used in this manual.

3 Worked Case: University Electricity Consumption

3.1 Case setting

Suppose a university wants to decide which electricity-related research question deserves institutional support. Instead of evaluating only one proposed topic, the committee constructs several candidate questions from the same declared evidence.

4 Step 1: Declare the Reality Space

Key idea

A research question must be evaluated against a fixed evidential domain. The Reality Space states exactly what the inquiry is trying to organise.

Why this step is necessary

DPT begins with a declared evidential domain so that a question is not evaluated in isolation from the observations that made it scientifically available. Without a fixed Reality Space, two evaluators may appear to score the same question while actually reasoning from different evidence.

Let the finite Reality Space be

$$R = \{r_1, r_2, \dots, r_8\}. \quad (1)$$

For the present case, the eight observations are listed in table 1.

Table 1: Declared Reality Space for the university electricity case.

Element	Observation or regularity
r_1	Electricity bills are increasing.
r_2	Classrooms sometimes remain powered when unoccupied.
r_3	Air-conditioning systems sometimes operate unnecessarily.
r_4	Solar panels generate power mainly during daylight hours.
r_5	Battery storage is costly.
r_6	Peak electricity demand commonly occurs in the afternoon.
r_7	Buildings show different consumption patterns.
r_8	Weather conditions affect cooling demand.

DPT permits observations, measurements, outcomes, anomalies, regularities, and theoretically articulated facts to serve as elements of a finite Reality Space [2].

Result and interpretation

The study now has a fixed object of inquiry. Every candidate question will be judged against the same eight observations rather than against an unspecified idea of “importance.”

The relationship among the eight declared observations is summarised visually in fig. 2.

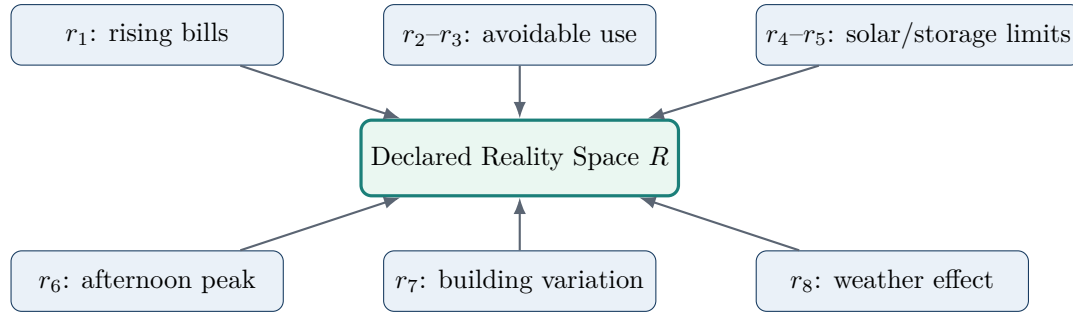


Figure 2: The eight observations are treated as one declared evidential domain before questions are compared.

5 Step 2: Declare the Background-Knowledge State

Key idea

Discovery Complexity is relative to what the team already knows and can already do. It is not an intrinsic property of a sentence.

Why this step is necessary

Discovery Complexity is background-dependent. A question that is difficult for a team without sensors, statistical knowledge, or energy expertise may be easier for a team that already possesses those capabilities. Declaring B prevents the score from being presented as an intrinsic property of the sentence alone.

Let the background-knowledge state be

$$B = \{b_1, b_2, b_3, b_4, b_5\}, \quad (2)$$

where the team has basic knowledge of electrical engineering, Internet of Things sensors, solar-energy systems, statistical forecasting, and university timetabling. The declared background knowledge and institutional capabilities are listed in table 2.

Table 2: Declared background knowledge and available institutional capability.

Element	Existing knowledge or capability	Why it matters
b_1	Electrical-energy fundamentals	Supports interpretation of consumption and demand.
b_2	IoT sensors	Makes occupancy and device-state measurement possible.
b_3	Solar-energy systems	Supports modelling of intermittent generation.
b_4	Statistical forecasting	Supports demand prediction and uncertainty analysis.
b_5	University timetables	Connects occupancy patterns with energy use.

6 Step 3: Generate Multiple Candidate Questions

Why this step is necessary

DPT is comparative. A single question can be assigned a coordinate, but dominance and frontier analysis require alternatives. Generating several questions also reduces the risk of accepting the first plausible topic proposed by a student or supervisor.

The committee proposes the four candidate questions presented in table 3.

Table 3: Four candidate questions generated from the same Reality Space.

Question	Informal type	Wording
q_1	Routine efficiency	Can replacing conventional bulbs with LED bulbs reduce university electricity consumption?
q_2	Organisational intervention	Can timetable changes reduce the university's peak electricity demand?
q_3	Integrated systems question	Can occupancy, timetable, weather, and solar-generation data be combined to predict and automatically minimise campus electricity demand?
q_4	Highly ambitious target	Can the campus become completely energy-self-sufficient throughout the year without drawing electricity from the grid?

All four questions will be evaluated under the same R , B , cost convention, description language, and normalising scales.

7 Step 4: Declare the Discovery-Cost Functional

Why this step is necessary

The formal definition of Discovery Complexity refers to the cost of an admissible procedure. Therefore, the study must state what “cost” means. If the cost rule is hidden or changed during evaluation, the resulting coordinates are not reproducible.

For classroom calculation, define six cost components, each on a 0–4 scale:

1. time requirement (T);
2. conceptual distance from existing knowledge (CD);
3. search or literature burden (S);
4. experimental burden (E);
5. computational burden (C); and
6. coordination burden (CO).

These components follow the operational structure reported in the computational benchmark [3]. The meaning of each ordinal score is declared in table 4.

Table 4: Declared 0–4 classroom scoring scale.

Score	Interpretation
0	Almost no additional requirement
1	Low requirement
2	Moderate requirement
3	High requirement
4	Very high requirement

For an operational procedure D , define

$$C(D) = T + CD + S + E + C + CO. \quad (3)$$

The maximum possible classroom cost is $6 \times 4 = 24$.

Important limitation

The 0–4 rubric is an operational proxy, not a universal physical unit. A real study should provide detailed anchors, independent coders, adjudication rules, reliability analysis, and sensitivity checks. The benchmark paper explicitly describes its components as deterministic proxies rather than direct universal estimators of the formal quantities [3].

8 Step 5: Identify Admissible Procedures and Calculate Discovery Complexity

Key idea

Discovery Complexity uses the least costly admissible route, not the average route and not the most elaborate route.

8.1 Formal definition

For question q , the feasible procedure set is

$$\mathcal{D}_q(R, B) = \{D \in \mathcal{D} : D(R, B) = q\}. \quad (4)$$

Discovery Complexity is

$$\text{DC}(q \mid R, B) = \begin{cases} \inf_{D \in \mathcal{D}_q(R, B)} C(D), & \mathcal{D}_q(R, B) \neq \emptyset, \\ +\infty, & \mathcal{D}_q(R, B) = \emptyset. \end{cases} \quad (5)$$

This means that among all admissible ways of making the question available, DPT takes the least cost. If no admissible procedure exists, the question is inaccessible under the declared conventions [2].

8.2 Manual component scores

The declared component scores for the four questions are reported in table 5.

Table 5: Manual Discovery Complexity component scores.

Component	q_1	q_2	q_3	q_4
Time (T)	1	2	3	4
Conceptual distance (CD)	1	2	3	4
Search (S)	1	2	3	4
Experiment (E)	1	2	3	4
Computation (C)	0	1	3	4
Coordination (CO)	1	3	3	4
Total cost	5	12	18	24

The normalised classroom score is

$$\widehat{DC}(q) = \frac{T + CD + S + E + C + CO}{24}. \quad (6)$$

Therefore,

$$\widehat{DC}(q_1) = \frac{5}{24} = 0.2083, \quad (7)$$

$$\widehat{DC}(q_2) = \frac{12}{24} = 0.5000, \quad (8)$$

$$\widehat{DC}(q_3) = \frac{18}{24} = 0.7500, \quad (9)$$

$$\widehat{DC}(q_4) = \frac{24}{24} = 1.0000. \quad (10)$$

8.3 Why the infimum matters

For q_3 , suppose the three admissible question-generation procedures reported in table 6 are identified.

Table 6: Alternative admissible procedures for generating q_3 .

Procedure	Description	Cost
D_1	Dean's observation followed by a faculty discussion	21
D_2	Literature review combined with a preliminary energy audit	18
D_3	Sensor pilot, audit, interdisciplinary workshop, and modelling exercise	20

Then

$$\mathcal{D}_{q_3}(R, B) = \{D_1, D_2, D_3\} \quad (11)$$

and

$$DC(q_3 \mid R, B) = \inf\{21, 18, 20\} = 18. \quad (12)$$

After normalisation,

$$\widehat{\text{DC}}(q_3) = \frac{18}{24} = 0.75. \quad (13)$$

Result and interpretation

The most elaborate procedure is not automatically used. The formal coordinate uses the least costly admissible route capable of generating the question. In this example, D_2 is the optimal observed procedure.

The complete sequence for constructing Discovery Complexity is summarised in fig. 3.

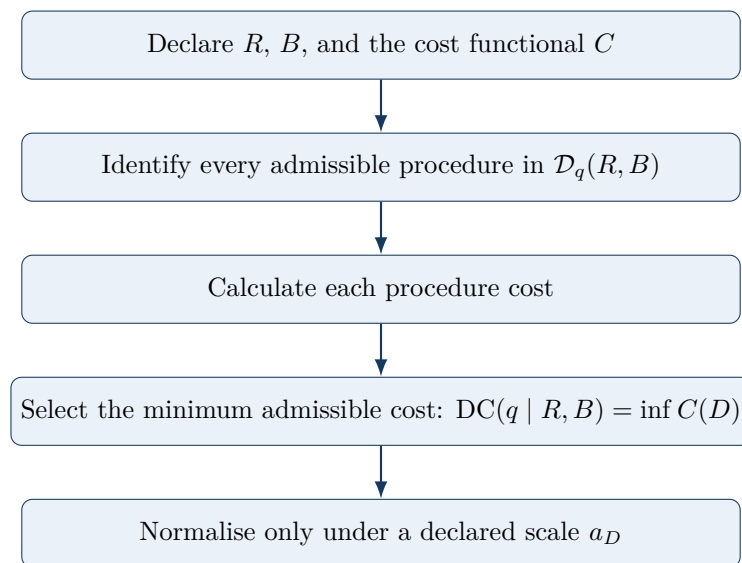


Figure 3: Manual workflow for constructing Discovery Complexity.

Try it yourself

Choose one question from your own discipline. List at least two genuinely different procedures through which that question could have become available. Define a cost rubric before assigning scores, then identify the least costly admissible route.

9 Step 6: Declare the Description-Length Functional

Why this step is necessary

Question Compression cannot be calculated until the study declares how representations are described. The phrase “this question explains many things” is too vague for reproducible analysis. The description language must be fixed before comparing questions.

For the classroom example, use *explanatory units*. An explanatory unit represents one coded proposition or necessary relational statement in a declared representation. Suppose that describing the eight observations separately requires

$$L(R) = 40 \text{ explanatory units.} \quad (14)$$

The value 40 is a transparent teaching convention. In a substantive study, L might be implemented using coded propositions, graph complexity, model description length, ontology statements, or another auditable representation.

10 Step 7: Transform the Reality Space Through Each Question

A question acts as a representation-transforming operator:

$$q(R) = R_q. \quad (15)$$

For each candidate, the evaluator constructs the reorganised representation and records its description length. The transformed representations and their resulting description lengths are summarised in table 7.

Table 7: Description length after each question reorganises the Reality Space.

Question	Main reorganisation induced by the question	$L(R_q)$
q_1	Connects lighting technology with one portion of electricity use, but leaves most observations separate.	35
q_2	Connects timetabling, occupancy, and peak demand, while leaving solar and weather relations partly separate.	25
q_3	Unifies occupancy, timetables, weather, solar supply, prediction, and automated control in one systems representation.	8
q_4	Organises the observations around self-sufficiency, but still requires several unresolved submodels for storage, seasonality, and demand.	20

11 Step 8: Calculate Signed Representational Change and Question Compression

Key idea

Question Compression asks how much shorter and more organised the declared representation becomes after the question restructures the evidence.

The signed representational change is

$$\Delta L(q \mid R) = L(R) - L(R_q). \quad (16)$$

Positive values indicate compression, zero indicates neutrality, and negative values indicate representational expansion. For a compression-explanatory question,

$$QC(q \mid R) = L(R) - L(R_q) = \Delta L(q \mid R) \geq 0. \quad (17)$$

These distinctions are important because a scientifically useful question can temporarily increase complexity; such expansion should not be mislabeled as compression [2].

Using $L(R) = 40$:

$$QC(q_1) = 40 - 35 = 5, \quad (18)$$

$$QC(q_2) = 40 - 25 = 15, \quad (19)$$

$$QC(q_3) = 40 - 8 = 32, \quad (20)$$

$$QC(q_4) = 40 - 20 = 20. \quad (21)$$

Declare the normalising scale $a_Q = 40$. Then

$$\widehat{QC}(q) = \frac{QC(q)}{40}. \quad (22)$$

Therefore,

$$\widehat{QC}(q_1) = \frac{5}{40} = 0.125, \quad (23)$$

$$\widehat{QC}(q_2) = \frac{15}{40} = 0.375, \quad (24)$$

$$\widehat{QC}(q_3) = \frac{32}{40} = 0.800, \quad (25)$$

$$\widehat{QC}(q_4) = \frac{20}{40} = 0.500. \quad (26)$$

Result and interpretation

Question q_3 has the largest Question Compression because it reorganises the eight observations into the shortest declared explanatory representation. Question q_4 sounds more ambitious, but it leaves more unresolved structure and therefore compresses less under the declared language.

The transition from the original representation to the transformed representation is illustrated in fig. 4.

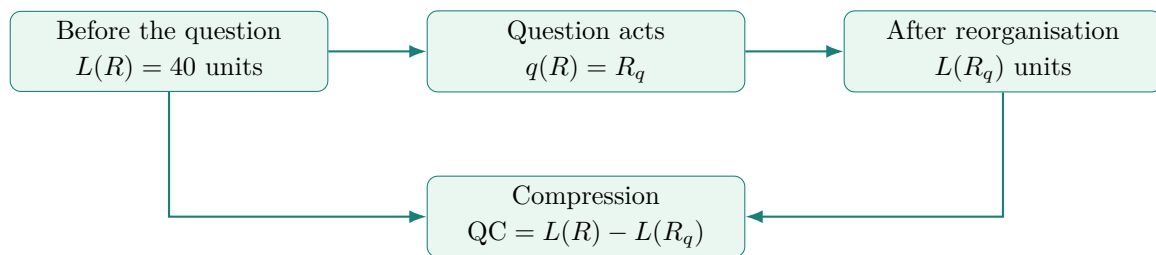


Figure 4: Question Compression compares the declared description length before and after a question reorganises the evidence.

12 Step 9: Construct the Discovery Coordinates

For every feasible compression-explanatory question, the Discovery coordinate is

$$\Pi(q \mid R, B) = (\text{DC}(q \mid R, B), \text{QC}(q \mid R)). \quad (27)$$

Using the declared scales $a_D = 24$ and $a_Q = 40$, the normalised coordinate is

$$\hat{\Pi}(q) = (\widehat{DC}(q), \widehat{QC}(q)). \quad (28)$$

The raw and normalised coordinates obtained from the preceding calculations are summarised in table 8.

Table 8: Final normalised coordinates used in the worked analysis.

Question	Raw DC	$\widehat{DC}$	Raw QC	$\widehat{QC}$
q_1	5	0.208	5	0.125
q_2	12	0.500	15	0.375
q_3	18	0.750	32	0.800
q_4	24	1.000	20	0.500

13 Step 10: Classify the Questions into Discovery Plane Regions

The benchmark used an exploratory threshold of 0.65 on both normalised coordinates [3]. For the worked example, declare

$$\tau_D = \tau_Q = 0.65. \quad (29)$$

The four descriptive region conditions are defined in table 9.

Table 9: The four descriptive Discovery Plane regions.

Region	DC condition	QC condition	Interpretation
I	$\widehat{DC} < 0.65$	$\widehat{QC} < 0.65$	Routine or locally reorganising
II	$\widehat{DC} < 0.65$	$\widehat{QC} \geq 0.65$	Efficient high-compression
III	$\widehat{DC} \geq 0.65$	$\widehat{QC} < 0.65$	Costly but weakly compressive
IV	$\widehat{DC} \geq 0.65$	$\widehat{QC} \geq 0.65$	Costly and strongly reorganising

Applying these conditions produces the classifications reported in table 10.

Table 10: Region classification of the four candidate questions.

Question	$\widehat{DC}$	$\widehat{QC}$	Region	Interpretation
q_1	0.208	0.125	I	Routine/local
q_2	0.500	0.375	I	Useful but locally reorganising
q_3	0.750	0.800	IV	Costly and strongly reorganising
q_4	1.000	0.500	III	Very costly but weakly compressive

The resulting positions of the four questions are plotted in fig. 5.

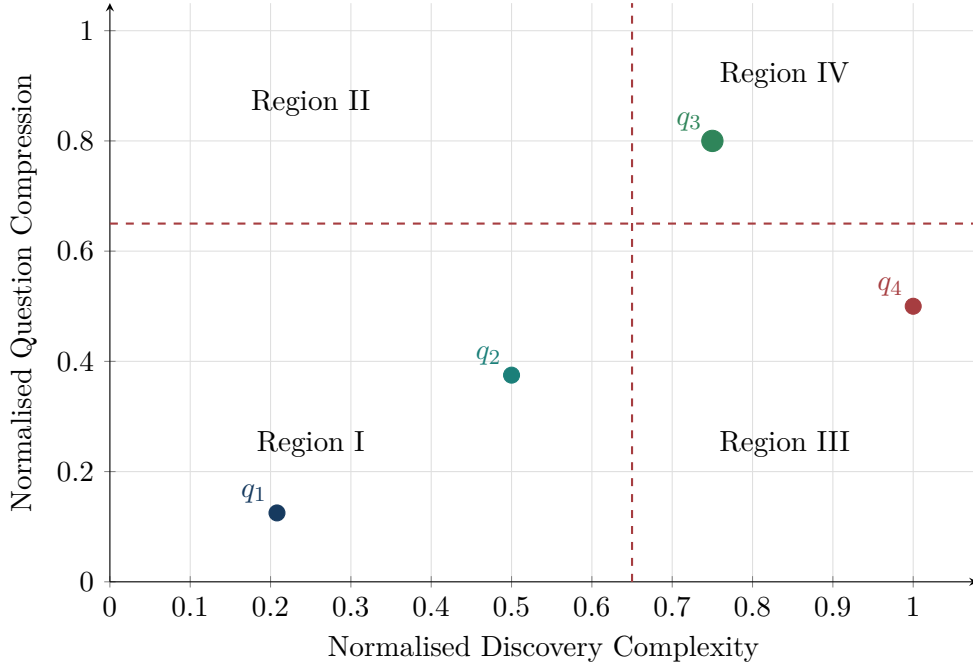


Figure 5: Discovery Plane for the four candidate questions under the declared 0.65 thresholds.

Important limitation

The regions are heuristic categories, not natural kinds. A threshold of 0.65 is an explicit analytical convention. The benchmark's sensitivity analysis showed that Region IV prevalence changes when the threshold changes [3].

14 Step 11: Verify Existence, Non-Negativity, and Bounds

Theorem – Existence and bounds

Meaning: A feasible compression-explanatory question has nonnegative coordinates, and its QC cannot exceed the original description length.

Why it matters: It provides a basic validity check before any comparison is attempted.

The existence-and-bounds result requires finite feasible DC and a compression-explanatory question. It implies

$$\Pi(q \mid R, B) \in \mathbb{R}_{\geq 0}^2, \quad 0 \leq \text{QC}(q \mid R) \leq L(R). \quad (30)$$

For q_3 :

$$\text{DC}(q_3) = 18 \geq 0 \quad (31)$$

and

$$0 \leq \text{QC}(q_3) = 32 \leq 40 = L(R). \quad (32)$$

Therefore, its coordinate satisfies the formal bounds [2].

15 Step 12: Verify Coordinate Uniqueness Under Fixed Conventions

Theorem – Coordinate uniqueness

Meaning: Once R , B , C , L , and the transformation are fixed, a feasible question has one coordinate.

Why it matters: Reproducibility requires that alternative scores arise only from explicitly changed conventions, not from hidden evaluator discretion.

Under fixed R , B , C , L , and deterministic transformation $q(R) = R_q$, each feasible compression-explanatory question has a unique coordinate [2]. For q_3 :

$$\hat{\Pi}(q_3) = (0.750, 0.800). \quad (33)$$

A different coordinate would be legitimate only if the study explicitly changed a convention, such as the background knowledge, cost rubric, representation language, normalising scale, or transformed representation.

16 Step 13: Verify Compression Monotonicity

Theorem – Compression monotonicity

Meaning: Under the same R and L , the question producing the shorter transformed description has the greater QC.

Why it matters: It makes comparative QC reasoning transparent and arithmetically checkable.

Compare q_3 and q_4 :

$$L(R_{q_3}) = 8 < 20 = L(R_{q_4}). \quad (34)$$

Because both are evaluated on the same R using the same L , compression monotonicity gives

$$\text{QC}(q_3) > \text{QC}(q_4). \quad (35)$$

Indeed,

$$32 > 20. \quad (36)$$

The shorter reorganised representation produces the larger Question Compression [2].

17 Step 14: Check Representation-Isomorphism Invariance

Suppose the Reality Space is translated from English into Hindi, or its observation labels are changed from r_i to s_i , while preserving all relations, admissible procedures, costs, and description lengths. If the mapping is an isomorphism preserving C and L , corresponding questions retain identical coordinates [2].

This step is a structural audit rather than a new numerical calculation. It prevents superficial changes in labels or language from producing different scientific evaluations.

18 Step 15: Calculate Dominance

Key idea

A question is stronger in the dominance sense only when it is no more costly and no less compressive than its competitor, with at least one strict advantage.

Question q_i dominates q_j when

$$\widehat{DC}(q_i) \leq \widehat{DC}(q_j), \quad \widehat{QC}(q_i) \geq \widehat{QC}(q_j), \quad (37)$$

with at least one strict inequality [2].

Compare q_3 with q_4 :

$$\widehat{DC}(q_3) = 0.750 < 1.000 = \widehat{DC}(q_4), \quad (38)$$

$$\widehat{QC}(q_3) = 0.800 > 0.500 = \widehat{QC}(q_4). \quad (39)$$

Therefore,

$$q_3 \succ q_4. \quad (40)$$

Result and interpretation

Question q_3 is both less costly and more knowledge-reorganising than q_4 . The more dramatic wording of q_4 does not make it the stronger candidate under the declared framework.

The geometric meaning of this dominance relationship is shown in fig. 6.

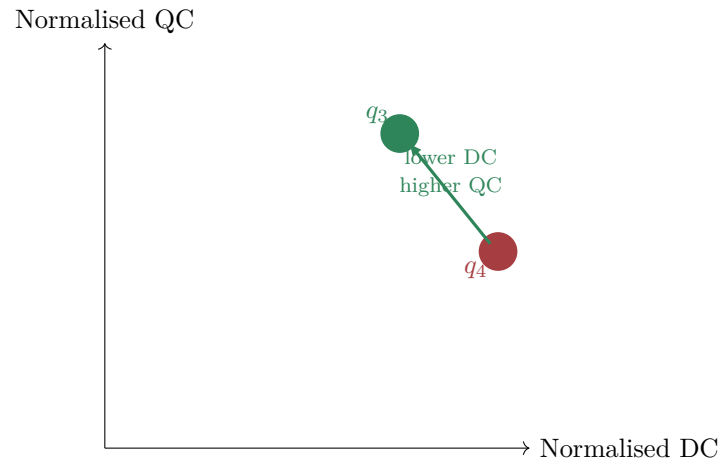


Figure 6: Dominance of q_4 by q_3 : the preferred direction is leftward in cost and upward in compression.

19 Step 16: Extract the Discovery Frontier

Key idea

The Discovery Frontier consists of all attainable questions that are *not dominated* by any other attainable question. Frontier membership therefore indicates nondominance, not scientific importance, correctness, ethical acceptability, or funding priority.

After computing the Discovery Plane coordinates and identifying dominance relationships, the next task is to determine the *Discovery Frontier*. For a finite set of questions ordered by increasing Discovery Complexity, this can be done manually using a simple scanning procedure.

Manual Frontier Extraction Procedure

1. Arrange all candidate questions in ascending order of $\widehat{DC}$.
2. Beginning with the first question, record the largest $\widehat{QC}$ observed so far.
3. Move from left to right through the ordered list.
4. Keep a question whenever its $\widehat{QC}$ is strictly greater than the maximum value previously encountered.
5. Otherwise, reject the question because it is dominated by an earlier question having lower or equal Discovery Complexity and greater or equal Question Compression.

For the present example, the questions are already ordered according to their Discovery Complexity:

$$q_1, q_2, q_3, q_4. \quad (41)$$

Their corresponding Question Compression values are

$$0.125, 0.375, 0.800, 0.500. \quad (42)$$

The frontier is extracted sequentially.

- Keep q_1 . It is the first candidate and therefore becomes the initial frontier point.
- Keep q_2 , since

$$0.375 > 0.125.$$
- Keep q_3 , since

$$0.800 > 0.375.$$
- Reject q_4 , since

$$0.500 < 0.800.$$

An earlier question (q_3) provides greater Question Compression while requiring lower Discovery Complexity.

Therefore, the Discovery Frontier is

$$\mathcal{F} = \{q_1, q_2, q_3\}. \quad (43)$$

Every non-empty finite attainable set contains at least one frontier-optimal question [2].

Important limitation

Frontier membership should not be interpreted as a recommendation to fund or pursue a research question. A routine low-complexity question may remain on the frontier simply because no cheaper attainable question offers greater Question Compression. In practice, research committees normally combine frontier analysis with additional criteria such as minimum Question Compression, originality, feasibility, ethical acceptability, resource availability, and expected scientific impact.

20 Step 17: Apply a Research-Budget Constraint

Suppose the institution can support only questions satisfying

$$\widehat{\text{DC}}(q) \leq \beta, \quad (44)$$

with $\beta = 0.80$. The feasible budget set is

$$F(0.80) = \{q : \widehat{\text{DC}}(q) \leq 0.80\} = \{q_1, q_2, q_3\}. \quad (45)$$

Their QC values are 0.125, 0.375, and 0.800. Therefore,

$$q^* = q_3. \quad (46)$$

The budget-frontier theorem states that a QC maximiser, with minimum DC used to break compression ties, is frontier-optimal relative to the finite budget set [2]. The combined effects of frontier membership and the declared budget are illustrated in fig. 7.

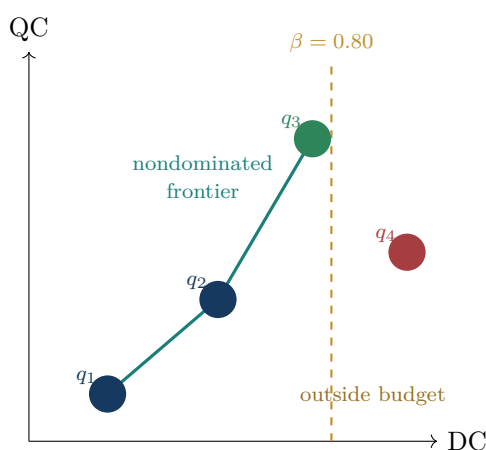


Figure 7: The frontier and the declared research budget jointly retain q_1 , q_2 , and q_3 , while excluding q_4 .

Try it yourself

Plot three or more candidate questions from your own field. Identify every nondominated point. Then impose a realistic resource budget and explain whether the frontier changes.

21 Step 18: Demonstrate Expansion of Procedure Sets

Initially, suppose only D_1 and D_3 are available:

$$\mathcal{D}_{q_3}^{(1)} = \{D_1, D_3\}, \quad \text{DC}^{(1)}(q_3) = \min\{21, 20\} = 20. \quad (47)$$

After an energy-audit capability is added, D_2 becomes available:

$$\mathcal{D}_{q_3}^{(2)} = \{D_1, D_2, D_3\}, \quad \text{DC}^{(2)}(q_3) = \min\{21, 18, 20\} = 18. \quad (48)$$

Because

$$\mathcal{D}_{q_3}^{(1)} \subseteq \mathcal{D}_{q_3}^{(2)}, \quad (49)$$

we obtain

$$\text{DC}^{(2)}(q_3) \leq \text{DC}^{(1)}(q_3). \quad (50)$$

Adding admissible procedures cannot increase the minimum cost when the cost convention is unchanged [2].

22 Step 19: Calculate Weighted Discovery Distance

For positive weights w_D, w_Q , the weighted discovery distance is

$$d_W(q_i, q_j) = \sqrt{w_D[\widehat{\text{DC}}(q_i) - \widehat{\text{DC}}(q_j)]^2 + w_Q[\widehat{\text{QC}}(q_i) - \widehat{\text{QC}}(q_j)]^2}. \quad (51)$$

Take $w_D = w_Q = 1$. Between q_3 and q_4 :

$$d_W(q_3, q_4) = \sqrt{(0.750 - 1.000)^2 + (0.800 - 0.500)^2} \quad (52)$$

$$= \sqrt{(-0.250)^2 + (0.300)^2} \quad (53)$$

$$= \sqrt{0.0625 + 0.0900} \quad (54)$$

$$= \sqrt{0.1525} \quad (55)$$

$$\approx 0.391. \quad (56)$$

This is coordinate distance, not proof of semantic similarity, causal dependence, or historical influence [2].

23 Step 20: Form a Neighbourhood and Calculate Path Length

For radius $\varepsilon > 0$,

$$N_\varepsilon(q) = \{q' : d_W(q, q') < \varepsilon\}. \quad (57)$$

Let $\varepsilon = 0.40$. Because

$$d_W(q_3, q_4) \approx 0.391 < 0.40, \quad (58)$$

we have

$$q_4 \in N_{0.40}(q_3). \quad (59)$$

They are close in coordinate space even though q_3 dominates q_4 .

For the ordered path

$$\gamma = (q_1, q_2, q_3), \quad (60)$$

path length is

$$\ell(\gamma) = d_W(q_1, q_2) + d_W(q_2, q_3). \quad (61)$$

First,

$$d_W(q_1, q_2) = \sqrt{(0.2083 - 0.5000)^2 + (0.125 - 0.375)^2} \quad (62)$$

$$\approx \sqrt{0.0851 + 0.0625} \quad (63)$$

$$\approx 0.384. \quad (64)$$

Next,

$$d_W(q_2, q_3) = \sqrt{(0.500 - 0.750)^2 + (0.375 - 0.800)^2} \quad (65)$$

$$= \sqrt{0.0625 + 0.180625} \quad (66)$$

$$\approx 0.493. \quad (67)$$

Thus,

$$\ell(\gamma) \approx 0.384 + 0.493 = 0.877. \quad (68)$$

24 Step 21: Construct an Admissibility Graph and Find a Geodesic

Coordinate distance alone does not determine whether one question can meaningfully develop into another. Define an admissibility graph whose edges represent historically, semantically, or procedurally acceptable transitions.

Assume the allowed edges and their lengths are:

Edge	Length
$q_1 \leftrightarrow q_2$	0.384
$q_2 \leftrightarrow q_3$	0.493
$q_2 \leftrightarrow q_4$	0.515
$q_3 \leftrightarrow q_4$	0.391

These admissible transitions and their associated lengths are represented in fig. 8.

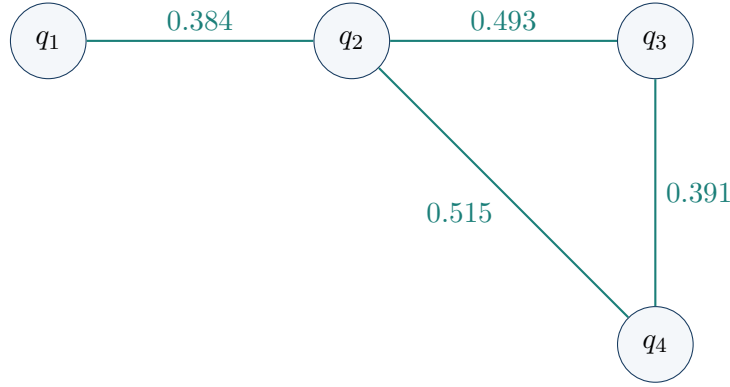


Figure 8: Illustrative admissibility graph for research-question development.

From q_1 to q_4 , two admissible paths are:

$$q_1 \rightarrow q_2 \rightarrow q_4 : 0.384 + 0.515 = 0.899, \quad (69)$$

$$q_1 \rightarrow q_2 \rightarrow q_3 \rightarrow q_4 : 0.384 + 0.493 + 0.391 = 1.268. \quad (70)$$

Therefore, the geodesic in this graph is

$$q_1 \rightarrow q_2 \rightarrow q_4 \quad (71)$$

with length 0.899. In a finite connected admissibility graph, a finite shortest path exists [2].

25 Step 22: Calculate Regularised Discovery Utility

A committee may require one scalar summary while still acknowledging the two-coordinate structure. For $\lambda > 0$, define

$$U_\lambda(q) = \widehat{\text{QC}}(q) - \lambda \widehat{\text{DC}}(q). \quad (72)$$

Let $\lambda = 0.50$. The resulting utility calculations are reported in table 11.

Table 11: Regularised discovery utility at $\lambda = 0.50$.

Question	Calculation	$U_{0.5}$
q_1	$0.125 - 0.5(0.2083)$	0.021
q_2	$0.375 - 0.5(0.5000)$	0.125
q_3	$0.800 - 0.5(0.7500)$	0.425
q_4	$0.500 - 0.5(1.0000)$	0.000

Therefore,

$$\arg \max_q U_{0.5}(q) = q_3. \quad (73)$$

Every utility maximiser is frontier-optimal, but a linear utility may fail to select every nondominated point when the attainable set is non-convex [2].

26 Step 23: Examine Non-Convexity and the Limitation of Scalarisation

The candidate coordinates form a discrete set. A point on a line segment between two attainable questions need not correspond to any admissible question. Therefore, the Discovery Frontier need not be convex [2].

For example, the midpoint between $q_2 = (0.500, 0.375)$ and $q_3 = (0.750, 0.800)$ is

$$\left(\frac{0.500 + 0.750}{2}, \frac{0.375 + 0.800}{2} \right) = (0.625, 0.5875). \quad (74)$$

No candidate question in the present set has this coordinate. It is a geometric interpolation, not automatically an attainable scientific question.

27 Step 24: Conduct Threshold Sensitivity Analysis

At $\tau_D = \tau_Q = 0.65$, q_3 is in Region IV. Raise both thresholds to 0.85:

$$\widehat{DC}(q_3) = 0.750 < 0.85, \quad (75)$$

while

$$\widehat{QC}(q_3) = 0.800 < 0.85. \quad (76)$$

It therefore moves to Region I under the mechanical four-region rule at the new thresholds. This does not mean the question itself changed; the classification convention changed. The classification of q_3 under alternative common thresholds is summarised in table 12.

Table 12: Region of q_3 under alternative common thresholds.

Common threshold	Coordinate comparison	Region of q_3
0.60	DC and QC both above threshold	IV
0.65	DC and QC both above threshold	IV
0.75	DC and QC both at or above threshold	IV
0.80	DC below, QC at threshold	II
0.85	DC and QC both below threshold	I

The benchmark varied the common threshold and reported that absolute Region IV prevalence depended materially on the convention [3].

28 Step 25: Conduct Weight Sensitivity Analysis

The equal-weight DC score treats all six cost dimensions identically. Suppose computation and coordination receive double weight:

$$\mathbf{w} = (1, 1, 1, 1, 2, 2). \quad (77)$$

For q_3 , whose component vector is $(3, 3, 3, 3, 3, 3)$,

$$\widehat{\text{DC}}_w(q_3) = \frac{1(3) + 1(3) + 1(3) + 1(3) + 2(3) + 2(3)}{4(1 + 1 + 1 + 1 + 2 + 2)} \quad (78)$$

$$= \frac{24}{32} = 0.750. \quad (79)$$

In this particular case, the value is unchanged because all six component scores are equal. For $q_2 = (2, 2, 2, 2, 1, 3)$,

$$\widehat{\text{DC}}_w(q_2) = \frac{2 + 2 + 2 + 2 + 2(1) + 2(3)}{32} \quad (80)$$

$$= \frac{16}{32} = 0.500. \quad (81)$$

Again, it remains unchanged. This is a useful teaching result: a sensitivity analysis can confirm stability rather than merely search for change.

To demonstrate a change, suppose a fifth candidate has component vector $(1, 1, 1, 1, 4, 4)$. Its equal-weight score is

$$\frac{12}{24} = 0.500, \quad (82)$$

but its reweighted score is

$$\frac{1 + 1 + 1 + 1 + 8 + 8}{32} = \frac{20}{32} = 0.625. \quad (83)$$

Thus, conclusions can depend on the institution's declared priorities. The benchmark used random component-weight draws to evaluate such sensitivity [3].

29 Step 26: Apply the Stability Bound Under Coordinate Error

Suppose estimated coordinate errors satisfy

$$|\widehat{\text{DC}}(q) - \text{DC}(q)| \leq \varepsilon_D, \quad |\widehat{\text{QC}}(q) - \text{QC}(q)| \leq \varepsilon_Q. \quad (84)$$

The DPT stability bound is

$$\left| \widehat{d}_W(q_i, q_j) - d_W(q_i, q_j) \right| \leq 2\sqrt{w_D \varepsilon_D^2 + w_Q \varepsilon_Q^2}. \quad (85)$$

Let $\varepsilon_D = \varepsilon_Q = 0.05$ and $w_D = w_Q = 1$. Then

$$\left| \widehat{d}_W - d_W \right| \leq 2\sqrt{0.05^2 + 0.05^2} \quad (86)$$

$$= 2\sqrt{0.005} \quad (87)$$

$$\approx 0.141. \quad (88)$$

Therefore, the estimated distance $d_W(q_3, q_4) \approx 0.391$ may differ from the underlying distance by no more than approximately 0.141 under the stated error assumptions [2].

30 Step 27: Construct a Dynamic Coordinate

DPT permits time-indexed conventions:

$$\Pi_t(q) = (\text{DC}(q \mid R_t, B_t), \text{QC}(q \mid R_t)). \quad (89)$$

Suppose that in 2026,

$$\hat{\Pi}_{2026}(q_3) = (0.750, 0.800). \quad (90)$$

After the university installs sensors and forms an interdisciplinary energy team, suppose the harmonised 2027 coordinate is

$$\hat{\Pi}_{2027}(q_3) = (0.600, 0.850). \quad (91)$$

The one-year finite changes are

$$\Delta \text{DC} = 0.600 - 0.750 = -0.150, \quad (92)$$

$$\Delta \text{QC} = 0.850 - 0.800 = 0.050. \quad (93)$$

The question has become less costly to generate and investigate, while its representational effect has increased.

Important limitation

A dynamic claim requires real longitudinal data and harmonised conventions. A coordinate change cannot by itself establish a paradigm shift. The formal paper explicitly separates the dynamic extension from claims not estimated by the available corpus [2].

The illustrative change in the coordinate of q_3 is shown in fig. 9.

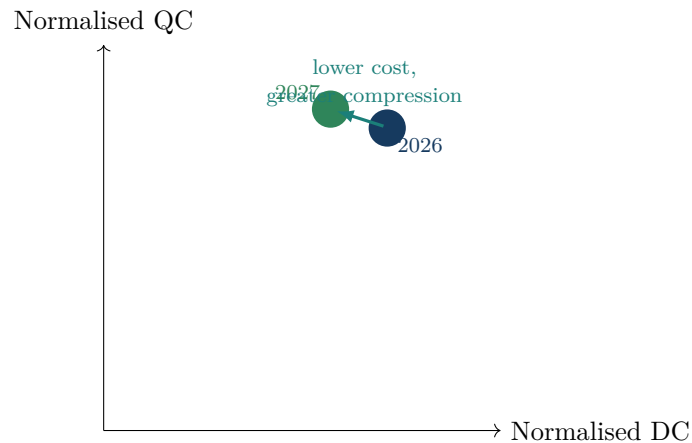


Figure 9: Illustrative dynamic movement of q_3 under harmonised conventions after new institutional capability is added.

31 Step 28: Demonstrate Threshold Scarcity

The four QC values are

$$0.125, 0.375, 0.800, 0.500. \quad (94)$$

At threshold $c_1 = 0.40$:

$$P\{\widehat{QC}(Q) \geq 0.40\} = \frac{2}{4} = 0.50. \quad (95)$$

At the higher threshold $c_2 = 0.70$:

$$P\{\widehat{QC}(Q) \geq 0.70\} = \frac{1}{4} = 0.25. \quad (96)$$

Therefore,

$$P\{\widehat{QC}(Q) \geq 0.70\} \leq P\{\widehat{QC}(Q) \geq 0.40\}. \quad (97)$$

This illustrates threshold scarcity: the event at the higher QC threshold is a subset of the event at the lower threshold [2].

It does not prove the stronger hypothesis that high-compression questions are systematically harder to generate. That claim requires a declared sampling process, a sufficiently large dataset, and statistical analysis of

$$E[DC(Q) \mid QC(Q) \geq c]. \quad (98)$$

32 Step 29: Relate the Manual Example to the Computational Benchmark

The benchmark analysed 1,000 AI-generated questions created from 100 seed problems across 20 scientific fields. Each seed received ten controlled reframing strategies. Six automated components operationalised DC, four operationalised QC, and questions were classified using declared thresholds. The benchmark reported 541 prospective Region IV questions and six nondominated frontier candidates under its primary convention [3].

The present manual follows the same broad logic but differs in scale and purpose, as summarised in table 13.

Table 13: Difference between the classroom manual and the computational benchmark.

Feature	Classroom manual	Published benchmark
Primary purpose	Teach every calculation transparently	Evaluate strategy effects across a large generated corpus
Number of questions	Four principal candidates	1,000 generated questions
Scoring	Declared pedagogical values	Archived automated proxy components
Statistics	Manual comparison and sensitivity illustrations	Repeated-measures tests, bootstrap intervals, and sensitivity analyses
Inference	Demonstration only	Computational benchmark evidence under declared conventions

33 Step 30: Make the Final Research-Screening Decision

Key idea

The final decision must be supported by the complete chain of declared evidence, calculations, comparisons, constraints, and limitations.

The evidence assembled in the preceding steps is summarised in table 14.

Table 14: Decision matrix for the four candidate questions.

Question	Region	Frontier	Dominated?	Budget-feasible?	Utility rank	Decision note
q_1	I	Yes	No	Yes	3	Low-cost routine question; useful for local improvement.
q_2	I	Yes	No	Yes	2	Moderately useful organisational study.
q_3	IV	Yes	No	Yes	1	Strongest balance of reorganising power and feasible complexity.
q_4	III	No	Yes, by q_3	No	4	Ambitious, but more costly and less compressive.

Result and interpretation**Selected candidate: q_3**

Can occupancy, timetable, weather, and solar-generation data be combined to predict and automatically minimise campus electricity demand?

Question q_3 is selected because it lies in Region IV, has the greatest QC, dominates q_4 , lies on the Discovery Frontier, satisfies the declared budget, and maximises regularised utility at $\lambda = 0.5$.

The complete decision sequence developed across the thirty steps is summarised in fig. 10.

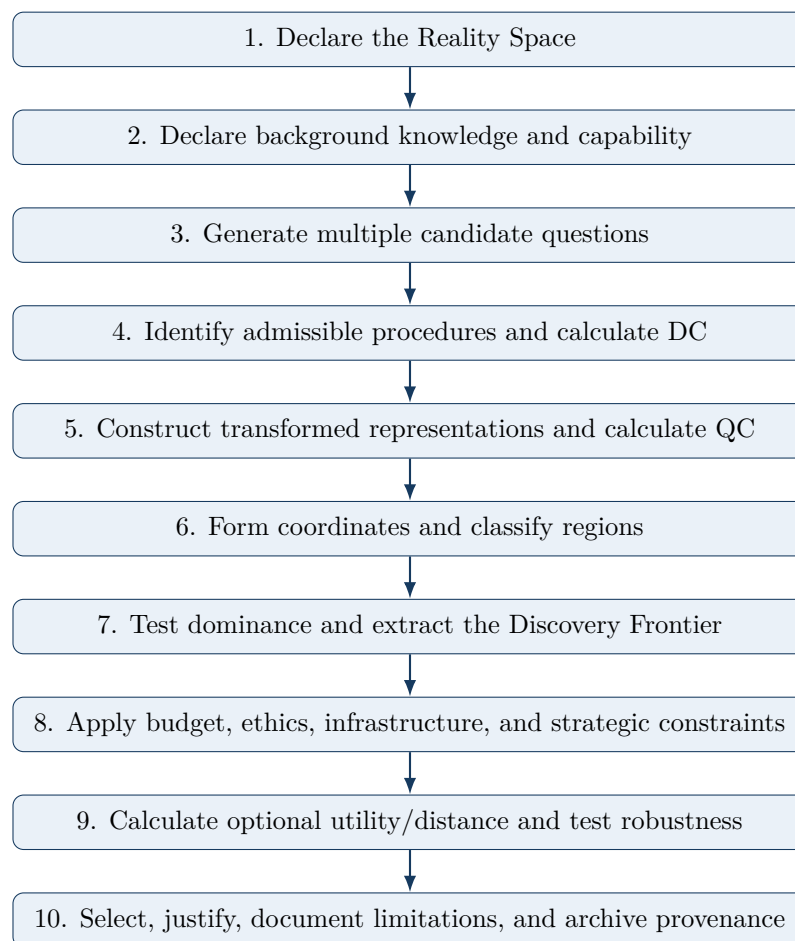


Figure 10: Complete DPT research-question screening workflow.

34 A Complete Student Worksheet

The complete reusable worksheet is provided in table 15. A student or research committee can reproduce the method by completing the following sequence.

Table 15: Reusable DPT worksheet for research-question screening.

Step	Required declaration or calculation	Record to be produced
1	Declare the Reality Space	Numbered observations, measurements, anomalies, or regularities.
2	Declare background knowledge	Existing theories, skills, data, tools, and institutional capabilities.
3	Generate alternatives	At least three candidate questions from the same evidence.
4	Fix the cost convention	Components, scales, weights, anchors, and coding rules.
5	Identify admissible procedures	Procedure set and cost of each procedure.
6	Calculate DC	Minimum admissible cost and normalised DC.
7	Fix the representation language	Definition of description length and coding unit.
8	Construct R_q	Reorganised representation induced by each question.
9	Calculate QC	$L(R) - L(R_q)$, signed change, and normalised QC.
10	Construct coordinates	$(\widehat{DC}, \widehat{QC})$ for every candidate.
11	Declare thresholds	τ_D and τ_Q , with justification.
12	Classify regions	Region I, II, III, or IV for each question.
13	Test dominance	Pairwise DC/QC comparison.
14	Extract frontier	Nondominated candidate set.
15	Apply constraints	Budget, ethics, time, infrastructure, or strategic limits.
16	Calculate distance/utility	Optional comparison metrics under declared weights.
17	Test robustness	Threshold, weight, representation, and error sensitivity.
18	Document limitations	Claims not established by the screening exercise.
19	Make decision	Selected question and transparent reasons.
20	Archive provenance	Rubric, coders, versions, data, calculations, and date.

35 Common Errors and How to Avoid Them

- Error 1: Scoring a sentence without declaring R and B .** The coordinate is representation- and background-relative.
- Error 2: Calling every difficult question important.** High DC with low QC belongs to Region III, not Region IV.
- Error 3: Treating a threshold as a natural law.** Thresholds are analytical conventions and require sensitivity analysis.
- Error 4: Using frontier membership as proof of major impact.** A low-cost routine question can be nondominated.
- Error 5: Changing rubrics after seeing the preferred result.** All conventions should be declared before final comparison.
- Error 6: Confusing compression with truth.** A question may organise a domain elegantly and still lead to a false explanation.
- Error 7: Ignoring uncertainty and coder disagreement.** Real applications need independent coding and reliability assessment.
- Error 8: Comparing unrelated disciplines without harmonisation.** Cross-domain coordinates require commensurable scales and anchor cases.
- Error 9: Assuming coordinate proximity means semantic similarity.** Distance is geometric unless an admissibility rule supplies additional meaning.
- Error 10: Reporting only the winning question.** Reproducibility requires the alternatives, scores, procedures, and rejected cases.

36 Frequently Asked Questions

Why use an infimum or minimum instead of an average?

Discovery Complexity asks for the least admissible cost of making a question available. An average would mix efficient and inefficient routes and would no longer represent the best available discovery procedure under the declared convention.

Can Question Compression be negative?

The signed change $\Delta L = L(R) - L(R_q)$ can be negative when the question expands the representation. The nonnegative QC coordinate is assigned only to compression-explanatory questions. Expansion can still be scientifically useful, but it should not be mislabeled as compression.

Does Region IV prove that a question will produce a discovery?

No. Region IV is a descriptive classification under declared thresholds. It does not establish truth, novelty, usefulness, ethics, feasibility, fundability, publication, or historical impact.

Why are the thresholds not universal?

Thresholds divide a continuous or discrete coordinate space for analysis. Their meaning depends on the declared scales, domain, coding procedure, and purpose. Robust studies report sensitivity to plausible alternatives.

Can two different questions have the same coordinate?

Yes. They are then coordinate-equivalent under the declared conventions. Equal coordinates do not imply identical wording, semantic content, or historical origin.

Can AI calculate DC and QC automatically?

AI can assist with coding and large-scale benchmarking, but the rubrics, evidence, admissible procedures, representation language, uncertainty, provenance, and validation rules must remain explicit and auditable.

Try it yourself

Apply the FAQ to your own candidate questions. State one conclusion your DPT screening can support and two conclusions it cannot support.

37 Formal Results: What This Example Demonstrates and What It Does Not

The formal results demonstrated by the worked example, together with the additional evidence required for substantive applications, are summarised in table 16.

Table 16: Coverage of formal DPT results in the worked example.

Formal result	Demonstrated in this manual	Additional evidence needed for a substantive study
Existence and bounds	Verified numerically for q_3 .	Valid cost and description functions.
Coordinate uniqueness	Explained under fixed conventions.	Deterministic transformation and fixed coding.
Compression monotonicity	Verified for q_3 and q_4 .	Same R and L across candidates.
Representation-isomorphism invariance	Illustrated through faithful labelling/translation.	Proven structure-preserving mapping.
Finite frontier existence	Frontier extracted from four points.	Complete finite attainable candidate set.
Budget-frontier theorem	Applied at $\beta = 0.80$.	Legitimate budget scale and tie rule.
Non-convexity	Illustrated through an unattained midpoint.	No mixture-closure assumption.
Expansion of procedure sets	Verified using two nested procedure sets.	Same cost functional before and after expansion.
Metric on coordinate classes	Distance calculated.	Harmonised coordinates and positive weights.
Dominance preorder	Used in pairwise comparison.	Coordinate equivalence handled explicitly.
Utility maximisers	q_3 maximises $U_{0.5}$.	Declared and justified λ .
Stability bound	Calculated for assumed error limits.	Empirical error bounds or uncertainty model.
Geodesic existence	Shortest path found in a finite graph.	Valid connected admissibility graph.
Convergence of nested frontiers	Not empirically established by four questions.	Increasing finite samples, compactness, density, and frontier approximation conditions.
Dynamic extension	Finite-change illustration provided.	Harmonised longitudinal data.
Threshold scarcity	Verified arithmetically in the sample.	Declared sampling process for probabilistic inference.
Strong discovery scarcity	Not established.	Large sample and appropriate statistical testing.

38 Conclusion

This report demonstrates a complete transition from observations to a transparent research-question decision. The selected question was not chosen because it sounded impressive or because a senior researcher preferred it. The process declared the Reality Space, background knowledge, admissible procedures, cost rule, description language, scales, thresholds, weights, and uncertainty assumptions. It then calculated coordinates, classified regions, tested dominance, extracted the frontier, imposed a budget, evaluated utility, and examined robustness.

The practical contribution of DPT can therefore be expressed simply:

**DPT converts “I believe this is a good question”
into “Here are the declared reasons, calculations,
comparisons, and limitations supporting this
question.”**

The framework does not replace expert judgement. It makes expert judgement more explicit, comparable, auditable, and reproducible. That is its value for students, supervisors, research committees, funding decisions, and AI-assisted scientific question generation.

A Compact Formula Sheet

The following formulas summarise the principal quantities developed throughout this guide. They are provided as a quick-reference sheet and use the notation established in the main text.

$$\begin{aligned}
R &= \{r_1, \dots, r_n\}, \\
q(R) &= R_q, \\
\mathcal{D}_q(R, B) &= \{D \in \mathcal{D} : D(R, B) = q\}, \\
\text{DC}(q \mid R, B) &= \begin{cases} \inf_{D \in \mathcal{D}_q(R, B)} C(D), & \mathcal{D}_q(R, B) \neq \emptyset, \\ +\infty, & \mathcal{D}_q(R, B) = \emptyset, \end{cases} \\
\Delta L(q \mid R) &= L(R) - L(R_q), \\
\text{QC}(q \mid R) &= L(R) - L(R_q) \geq 0, \quad \text{for a compression-explanatory question,} \\
\Pi(q \mid R, B) &= (\text{DC}(q \mid R, B), \text{QC}(q \mid R)), \\
\widehat{\text{DC}}(q) &= \frac{\text{DC}(q)}{a_D}, \\
\widehat{\text{QC}}(q) &= \frac{\text{QC}(q)}{a_Q}, \\
q_i \succ q_j &\iff \text{DC}(q_i) \leq \text{DC}(q_j), \quad \text{QC}(q_i) \geq \text{QC}(q_j), \quad \text{with at least one strict inequality,} \\
d_W(q_i, q_j) &= \sqrt{w_D[\text{DC}(q_i) - \text{DC}(q_j)]^2 + w_Q[\text{QC}(q_i) - \text{QC}(q_j)]^2}, \\
N_\varepsilon(q) &= \{q' : d_W(q, q') < \varepsilon\}, \\
\ell(\gamma) &= \sum_{k=0}^{m-1} d_W(q_k, q_{k+1}), \\
U_\lambda(q) &= \text{QC}(q) - \lambda \text{DC}(q), \\
\Pi_t(q) &= (\text{DC}(q \mid R_t, B_t), \text{QC}(q \mid R_t)).
\end{aligned}$$

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